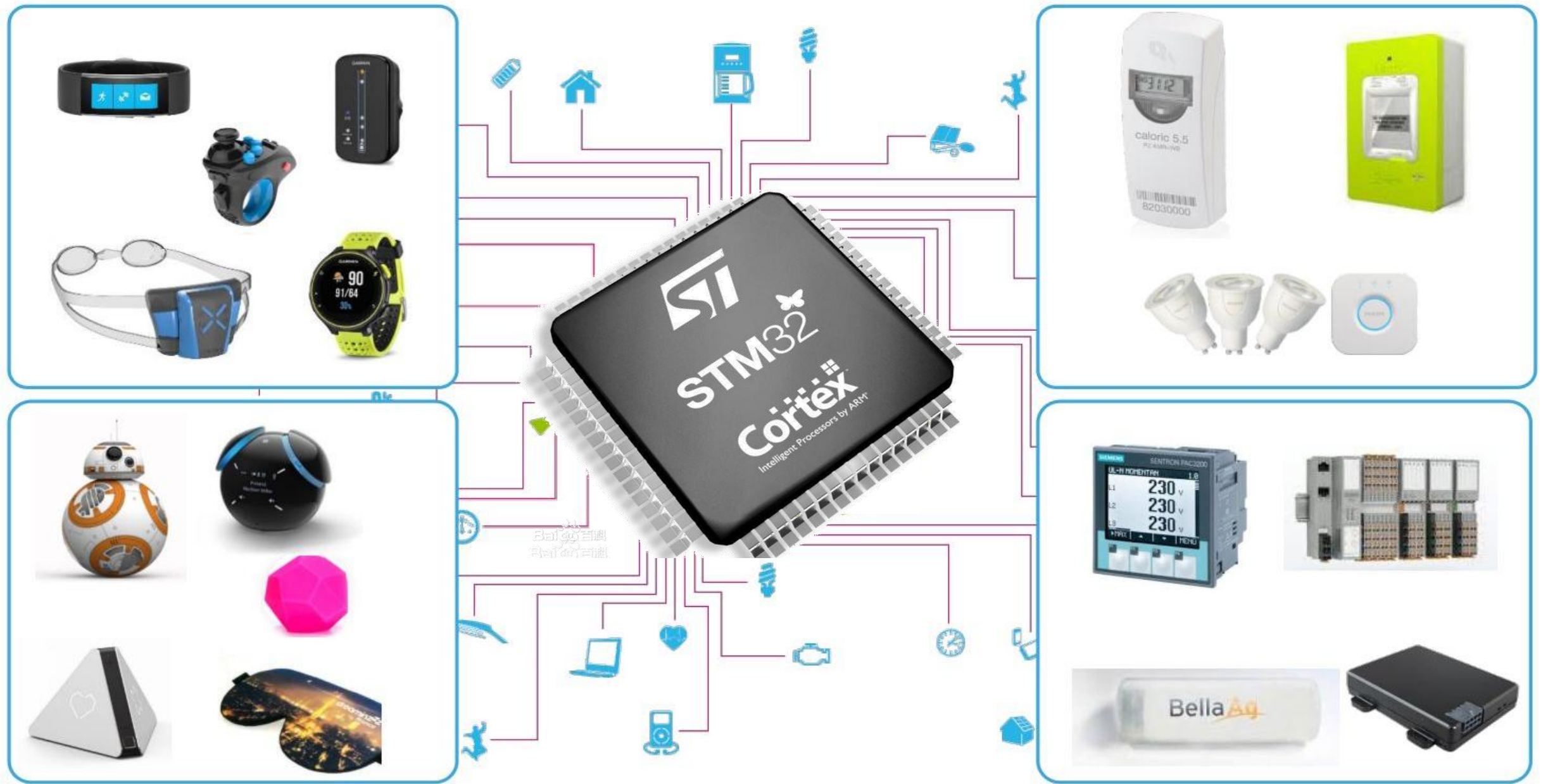




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Lecture8

Timer-III

Topics

- Overview of Advanced Timer
- Advanced Timer--Complementary PWM with
Deadtime Insertion and Break

Part I : Overview of Advanced Timer

14.1 TIM1 and TIM8 introduction

The advanced-control timers (TIM1 and TIM8) consist of a 16-bit auto-reload counter driven by a programmable prescaler.

It may be used for a variety of purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

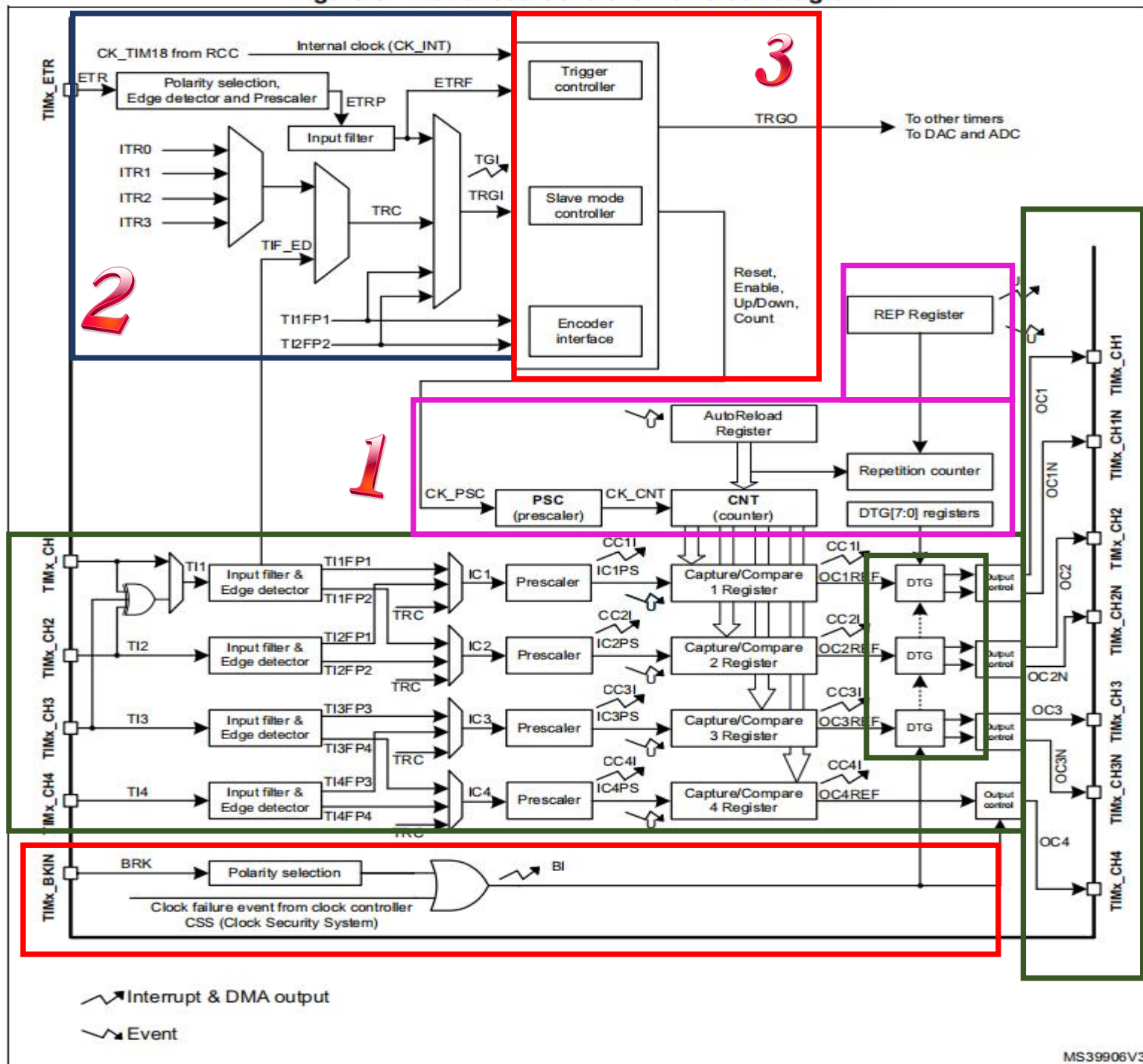
Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

The advanced-control (TIM1 and TIM8) and general-purpose (TIMx) timers are completely independent, and do not share any resources. They can be synchronized together as described in [Section 14.3.20](#).

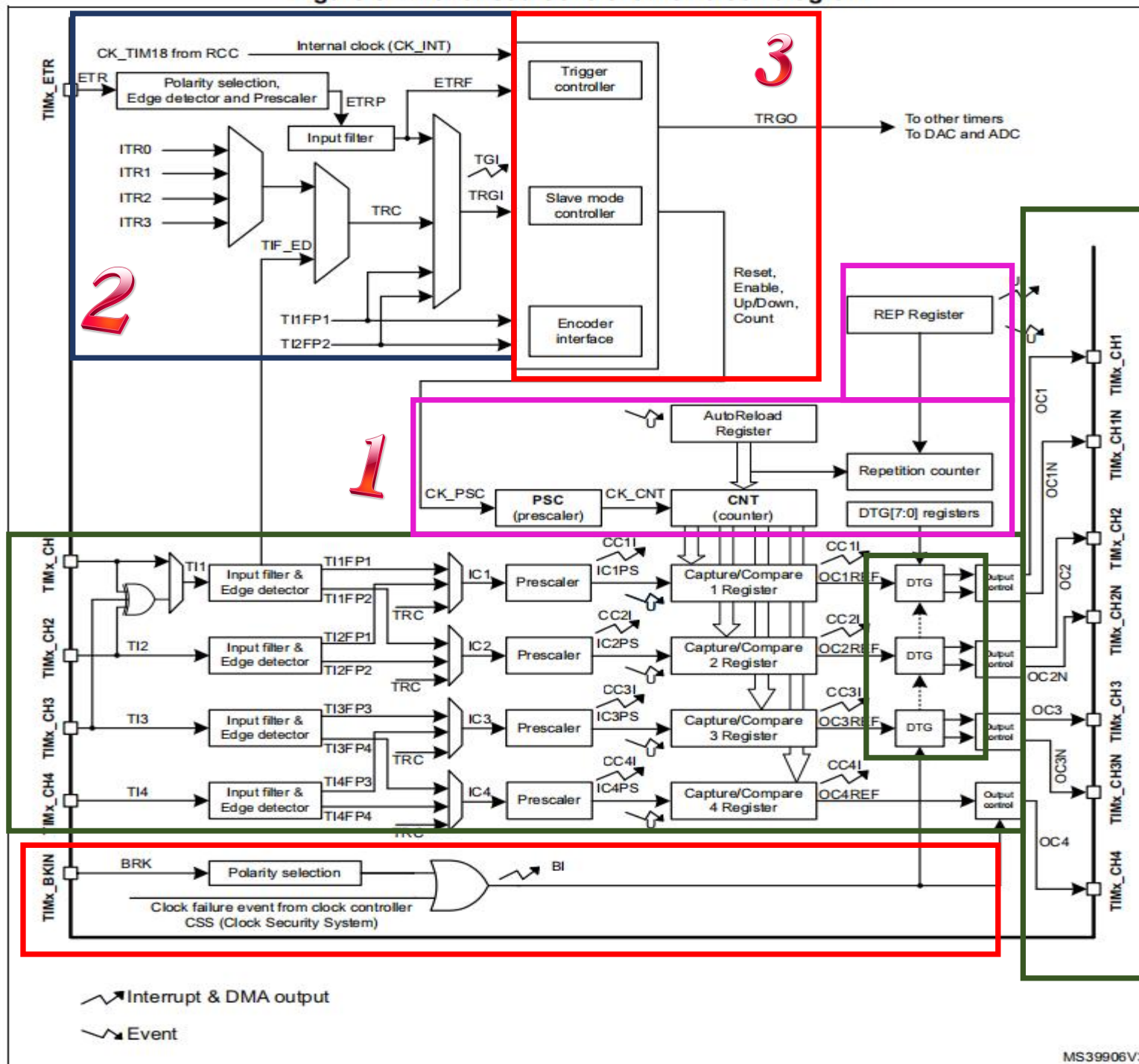
14.2 TIM1 and TIM8 main features

TIM1 and TIM8 timer features include:

- 16-bit up, down, up/down auto-reload counter.
- 16-bit programmable prescaler allowing dividing (also “on the fly”) the counter clock frequency either by any factor between 1 and 65536.
- Up to 4 independent channels for:
 - Input capture
 - Output compare
 - PWM generation (Edge and Center-aligned Mode)
 - One-pulse mode output
- Complementary outputs with programmable dead-time
- Synchronization circuit to control the timer with external signals and to interconnect several timers together.
- Repetition counter to update the timer registers only after a given number of cycles of the counter.
- Break input to put the timer’s output signals in reset state or in a known state.
- Interrupt/DMA generation on the following events:
 - Update: counter overflow/underflow, counter initialization (by software or internal/external trigger)
 - Trigger event (counter start, stop, initialization or count by internal/external trigger)
 - Input capture
 - Output compare
 - Break input
- Supports incremental (quadrature) encoder and hall-sensor circuitry for positioning purposes
- Trigger input for external clock or cycle-by-cycle current management



- 1
 - 14.3.1 Time-base unit
 - 14.3.2 Counter modes
 - Upcounting mode
 - Downcounting mode
 - Center-aligned mode (up/down counting)
 - 14.3.3 Repetition counter
- 2
 - 14.3.4 Clock selection
 - Internal clock source (CK_INT)
 - External clock source mode 1
 - External clock source mode 2
- 4
 - 14.3.5 Capture/compare channels
 - 14.3.6 Input capture mode
 - 14.3.7 PWM input mode
 - 14.3.8 Forced output mode
 - 14.3.9 Output compare mode
- 5
 - 14.3.10 PWM mode
 - 14.3.11 Complementary outputs and dead-time insertion
 - 14.3.12 Using the break function
 - 14.3.13 Clearing the OCxREF signal on an external event
 - 14.3.14 6-step PWM generation
 - 14.3.15 One-pulse mode
 - 14.3.16 Encoder interface mode
 - 14.3.17 Timer input XOR function
 - 14.3.18 Interfacing with Hall sensors
 - 14.3.19 TIMx and external trigger synchronization
 - 14.3.20 Timer synchronization
 - 14.3.21 Debug mode



14.4 TIM1 and TIM8 registers

14.4.1 TIM1 and TIM8 control register 1 (TIMx_CR1)

14.4.2 TIM1 and TIM8 control register 2 (TIMx_CR2)

14.4.3 TIM1 and TIM8 slave mode control register (TIMx_SMCR)

14.4.4 TIM1 and TIM8 DMA/interrupt enable register (TIMx_DIER)

14.4.5 TIM1 and TIM8 status register (TIMx_SR)

14.4.6 TIM1 and TIM8 event generation register (TIMx_EGR)

14.4.7 TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1)

14.4.8 TIM1 and TIM8 capture/compare mode register 2 (TIMx_CCMR2)

14.4.9 TIM1 and TIM8 capture/compare enable register (TIMx_CCER)

14.4.10 TIM1 and TIM8 counter (TIMx_CNT)

14.4.11 TIM1 and TIM8 prescaler (TIMx_PSC)

14.4.12 TIM1 and TIM8 auto-reload register (TIMx_ARR)

14.4.13 TIM1 and TIM8 repetition counter register (TIMx_RCR)

14.4.14 TIM1 and TIM8 capture/compare register 1 (TIMx_CCR1)

14.4.15 TIM1 and TIM8 capture/compare register 2 (TIMx_CCR2)

14.4.16 TIM1 and TIM8 capture/compare register 3 (TIMx_CCR3)

14.4.17 TIM1 and TIM8 capture/compare register 4 (TIMx_CCR4)

14.4.18 TIM1 and TIM8 break and dead-time register (TIMx_BDTR)

14.4.19 TIM1 and TIM8 DMA control register (TIMx_DCR)

14.4.20 TIM1 and TIM8 DMA address for full transfer (TIMx_DMAR)

14.3.3 Repetition counter

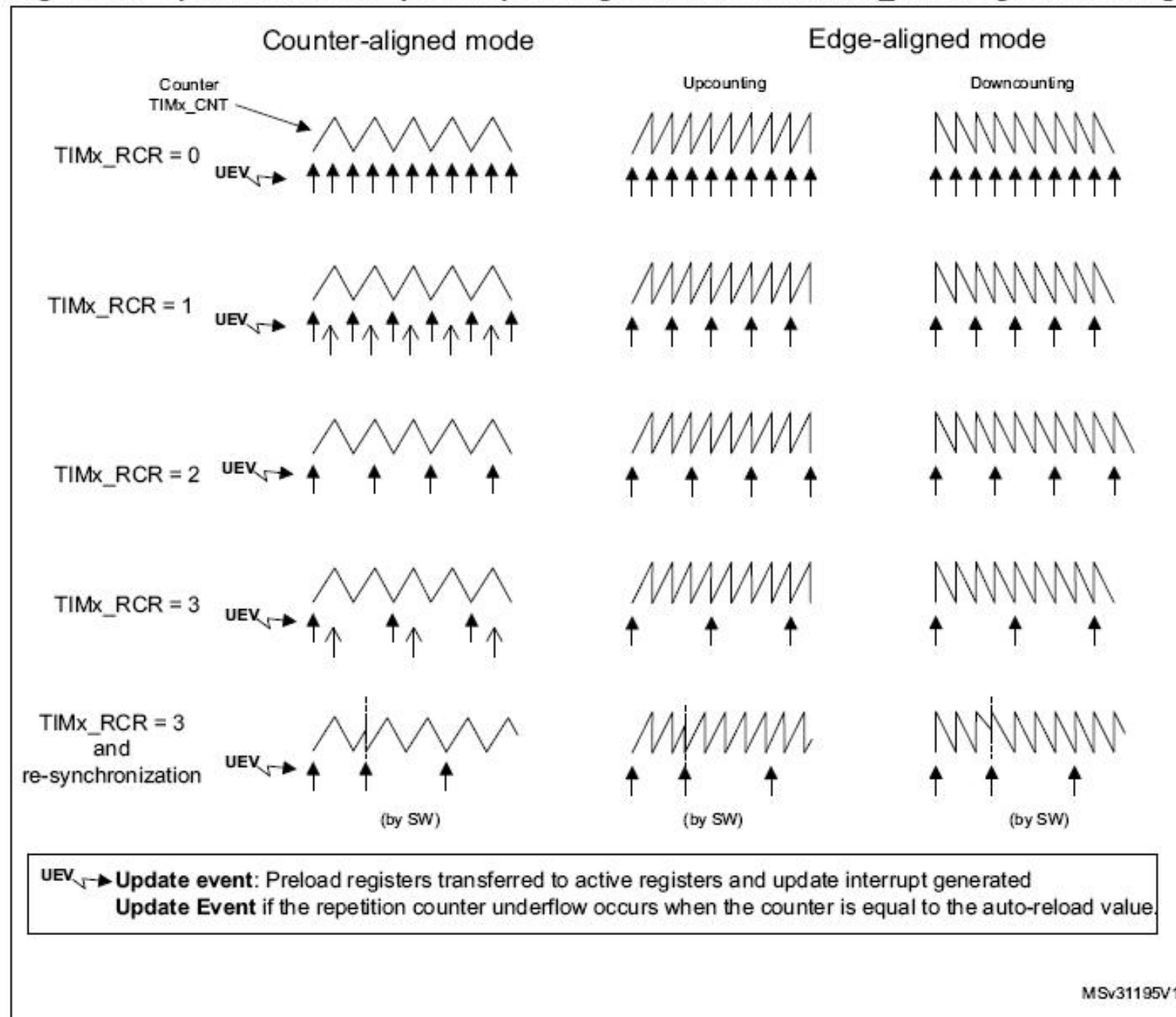
Section 14.3.1: Time-base unit describes how the update event (UEV) is generated with respect to the counter overflows/underflows. It is actually generated only when the repetition counter has reached zero. This can be useful when generating PWM signals.

This means that data are transferred from the preload registers to the shadow registers (TIMx_ARR auto-reload register, TIMx_PSC prescaler register, but also TIMx_CCRx capture/compare registers in compare mode) every N+1 counter overflows or underflows, where N is the value in the TIMx_RCR repetition counter register.

The repetition counter is decremented:

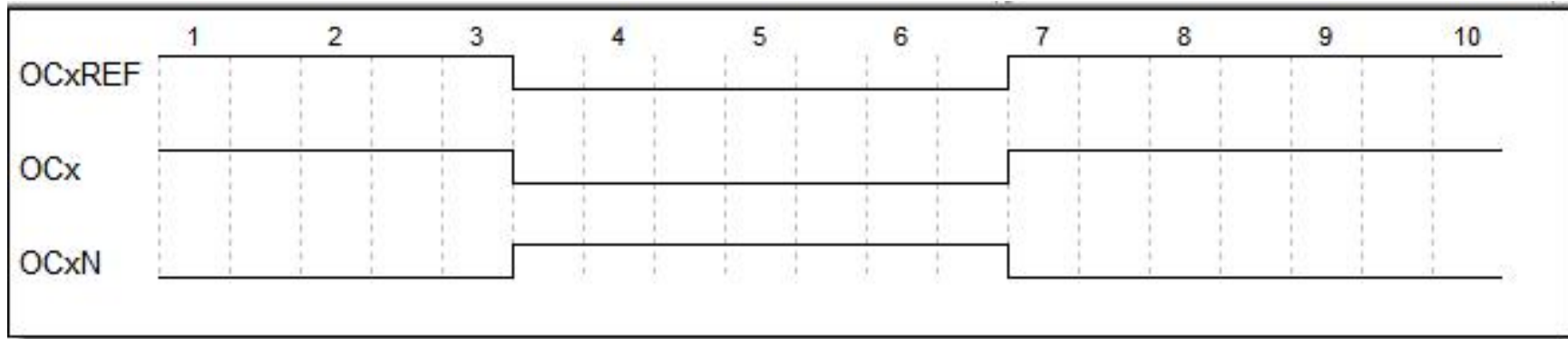
- At each counter overflow in upcounting mode,
 - At each counter underflow in downcounting mode,
 - At each counter overflow and at each counter underflow in center-aligned mode.
- Although this limits the maximum number of repetition to 128 PWM cycles, it makes it possible to update the duty cycle twice per PWM period. When refreshing compare registers only once per PWM period in center-aligned mode, maximum resolution is $2 \times T_{ck}$, due to the symmetry of the pattern.

Figure 72. Update rate examples depending on mode and TIMx_RCR register settings



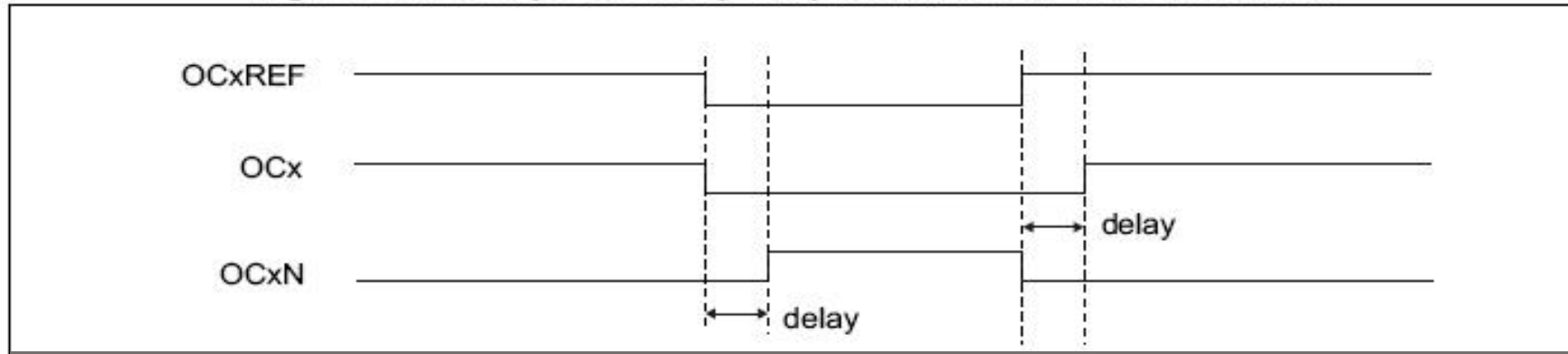
➤ Part II : Advanced Timer--Complementary PWM with
Deadtime Insertion and Break

Complementary PWM

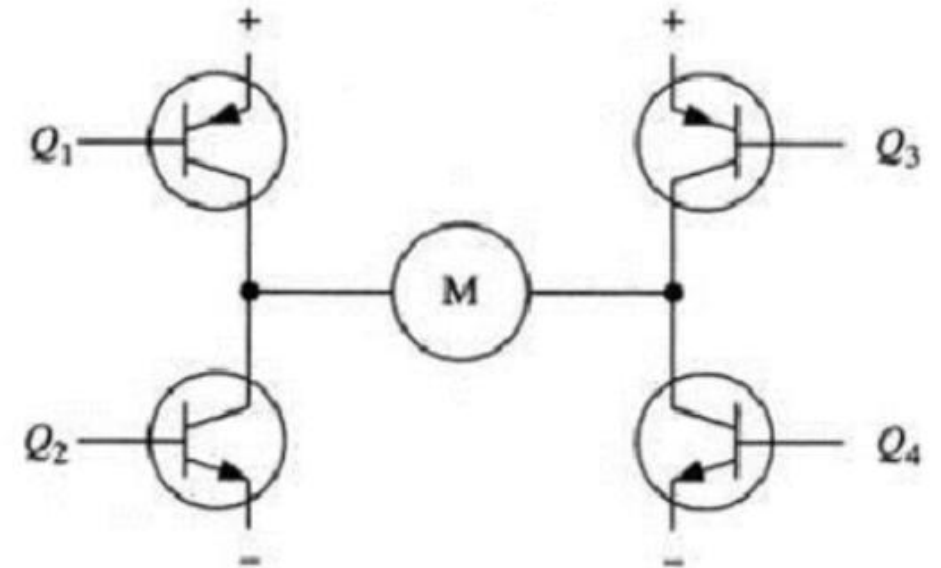
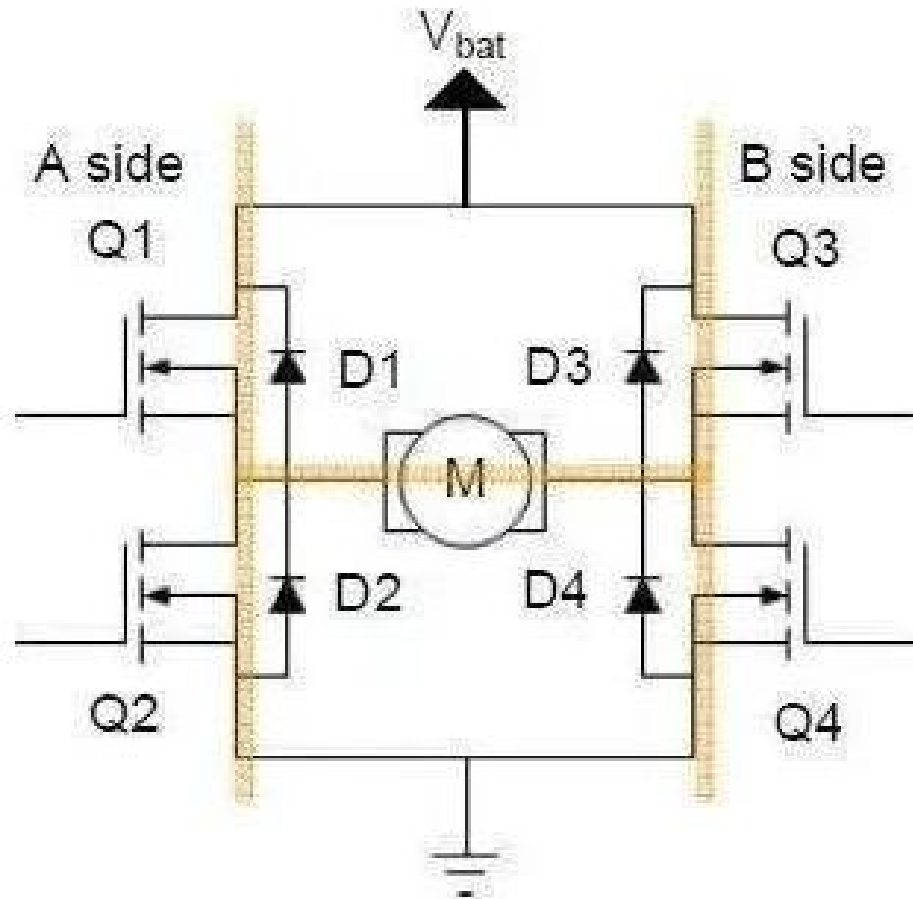


Complementary PWM with Deadtime Insertion

Figure 86. Complementary output with dead-time insertion.

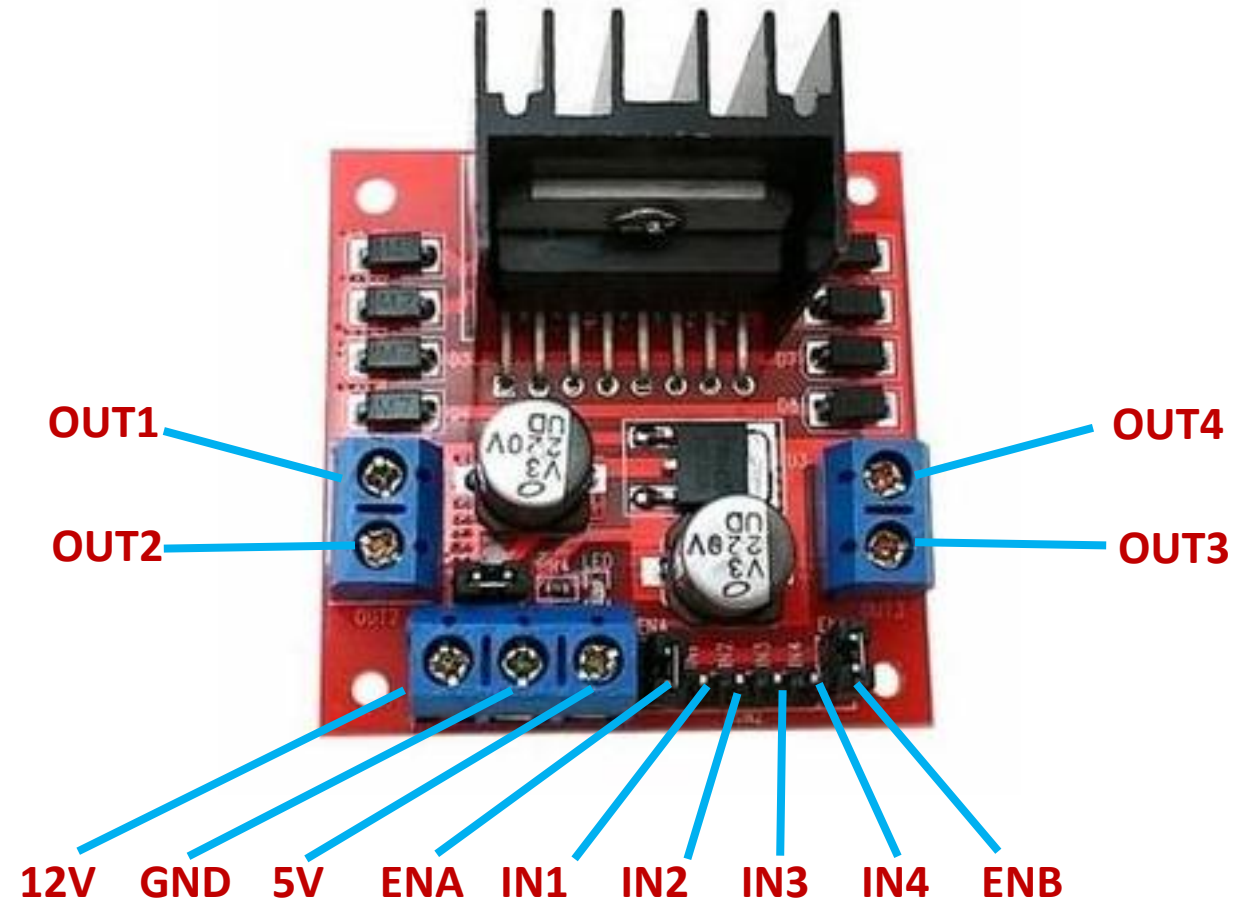


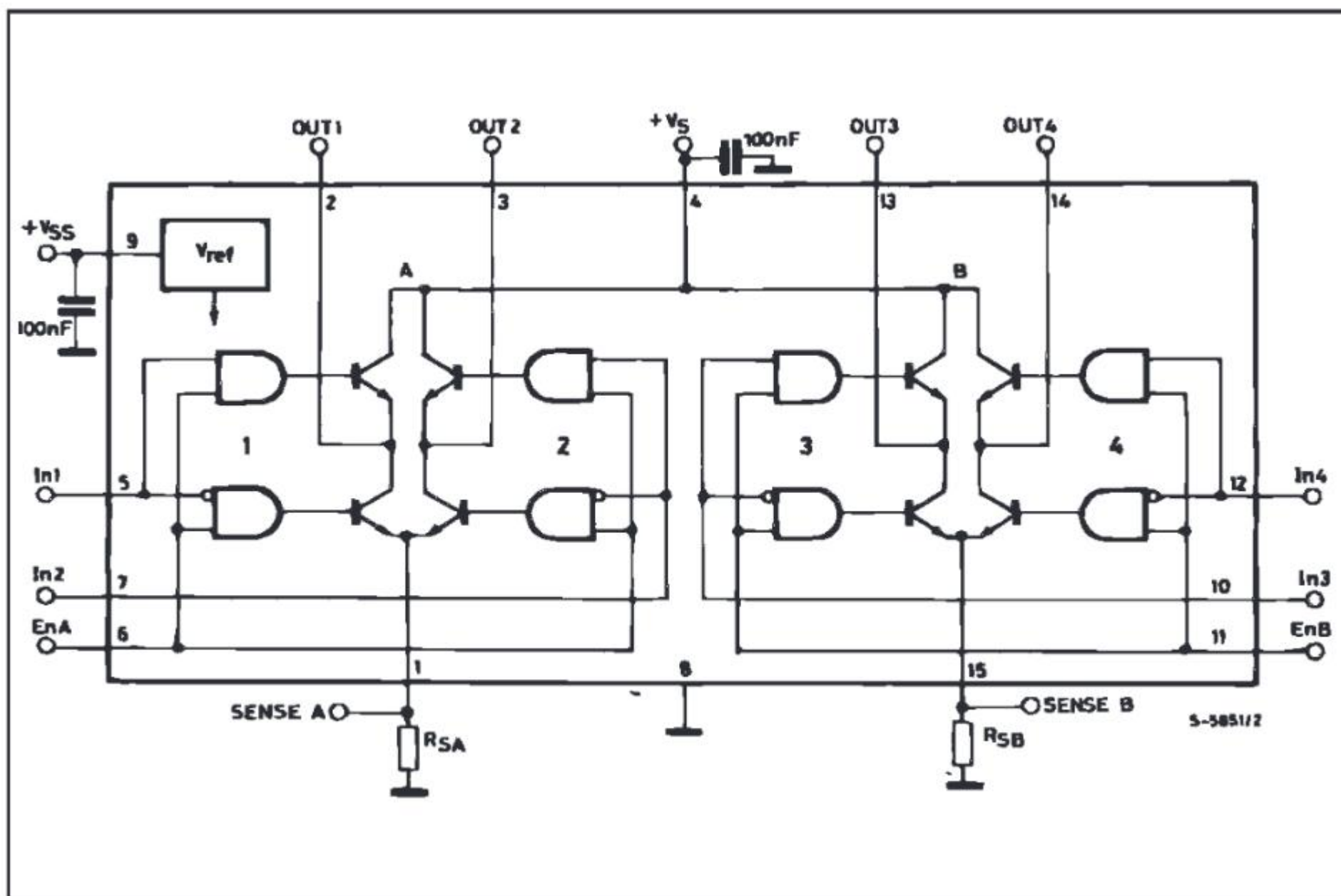
Why Deadtime Insertion

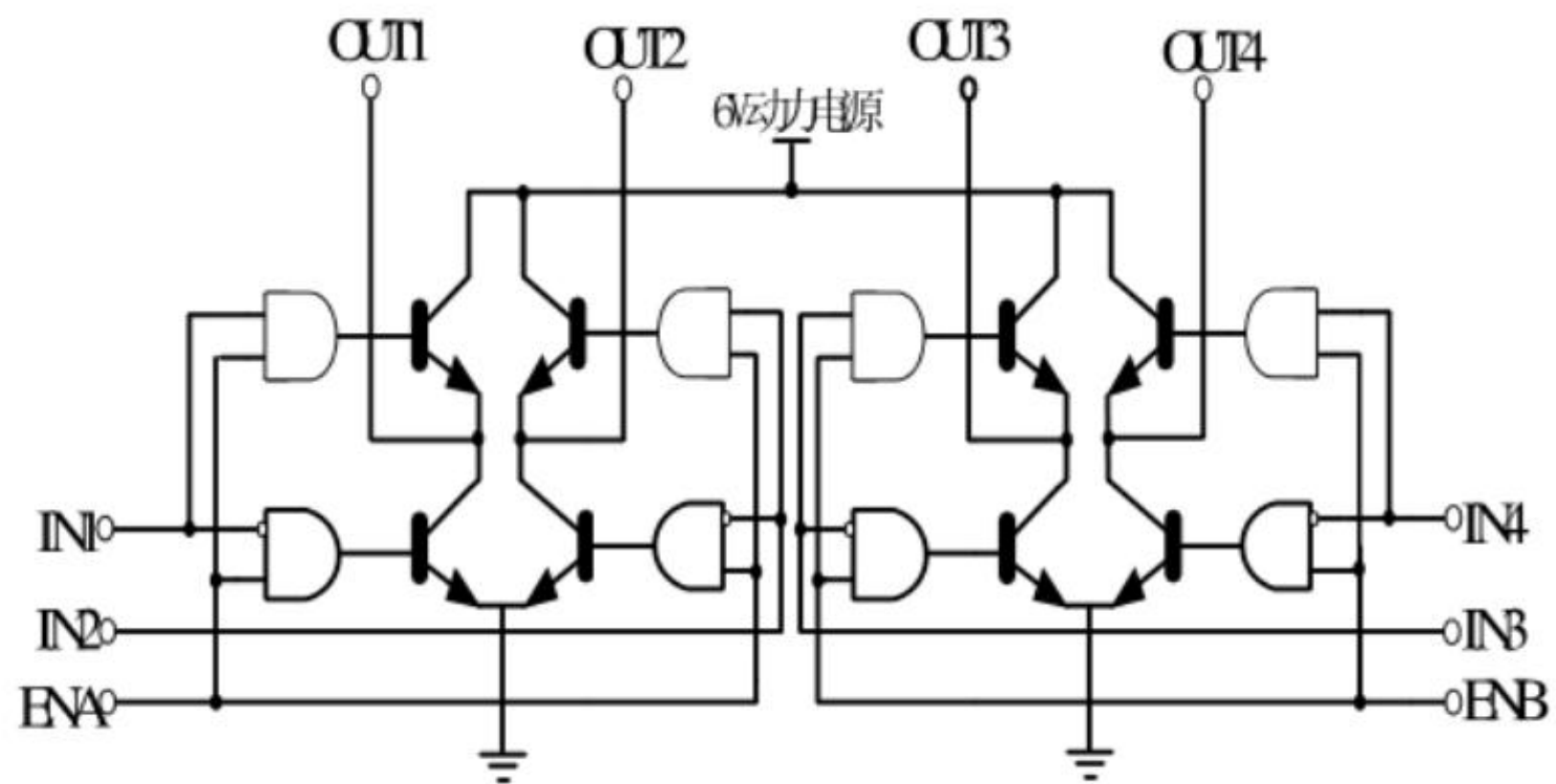


$\left\{ \begin{array}{l} Q_1, Q_4 \text{ on and } Q_2, Q_3 \text{ off} \\ Q_1, Q_4 \text{ off and } Q_2, Q_3 \text{ on} \end{array} \right.$

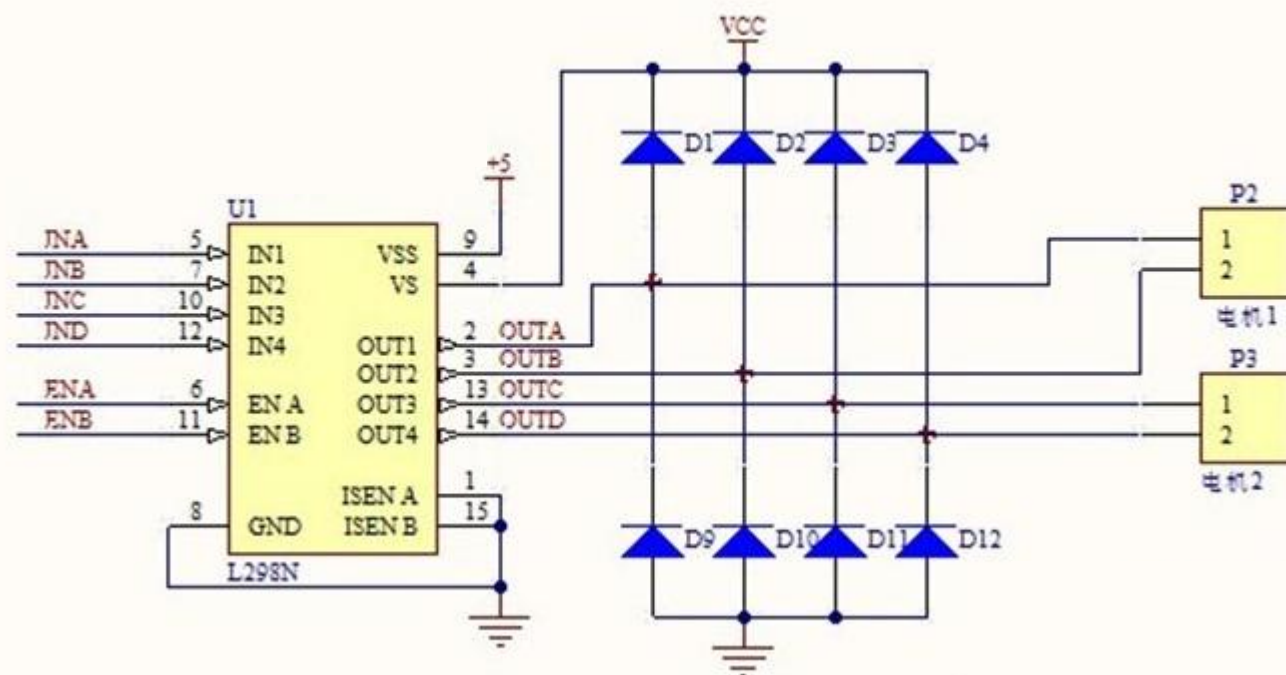
L298N







引脚名称	引脚标号	I/O	功能	引脚名称	引脚标号	I/O	功能
Sense A	1	O	H桥A的电流控制，可通过一个电阻接地限制电流(不限电流时直接接地)	VSS	9	—	给内部逻辑电路供电，一般接5V
Out 1	2	O	H桥A的输出1脚	Input 3	10	I	H桥B的逻辑输入1
Out 2	3	O	H桥A的输出2脚	Enable B	11	I	H桥B的使能控制端，高电平打开，低电平关闭
Vs	4	—	电机驱动电压3~48V，需要一个100nF的滤波电容接地	Input 4	12	I	H桥B的逻辑输入2
Input 1	5	I	H桥A的逻辑输入1	Out 3	13	O	H桥B的输出1脚
Enable A	6	I	H桥A的使能控制端，高电平打开，低电平关闭	Out 4	14	O	H桥B的输出2脚
Input 2	7	I	H桥A的逻辑输入2	Sense B	15	O	H桥B的电流控制，可通过一个电阻接地限制电流(不限电流时直接接地)
GND	8	—	接地				



ENA	IN1	IN2	直流电机状态
0	X	X	停止
1	0	0	制动
1	0	1	正转
1	1	0	反转
1	1	1	制动

How to Realize Deadtime Insertion

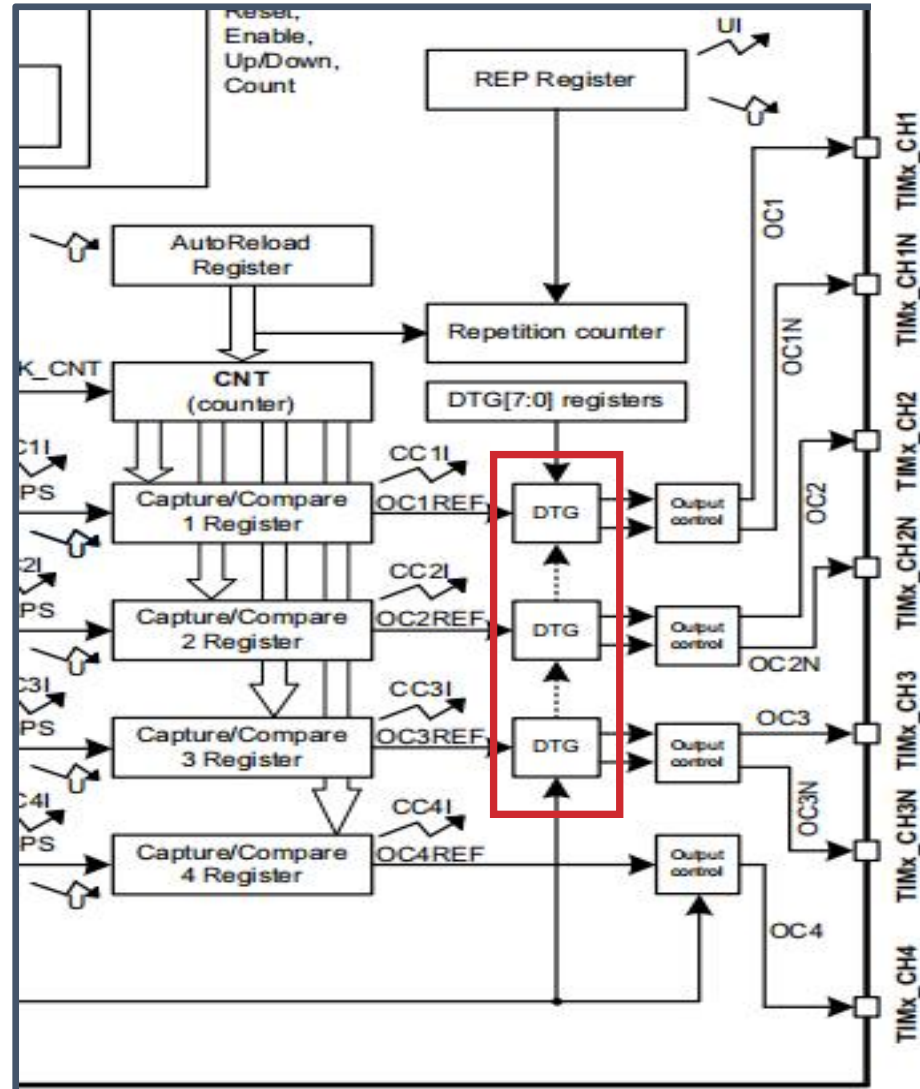
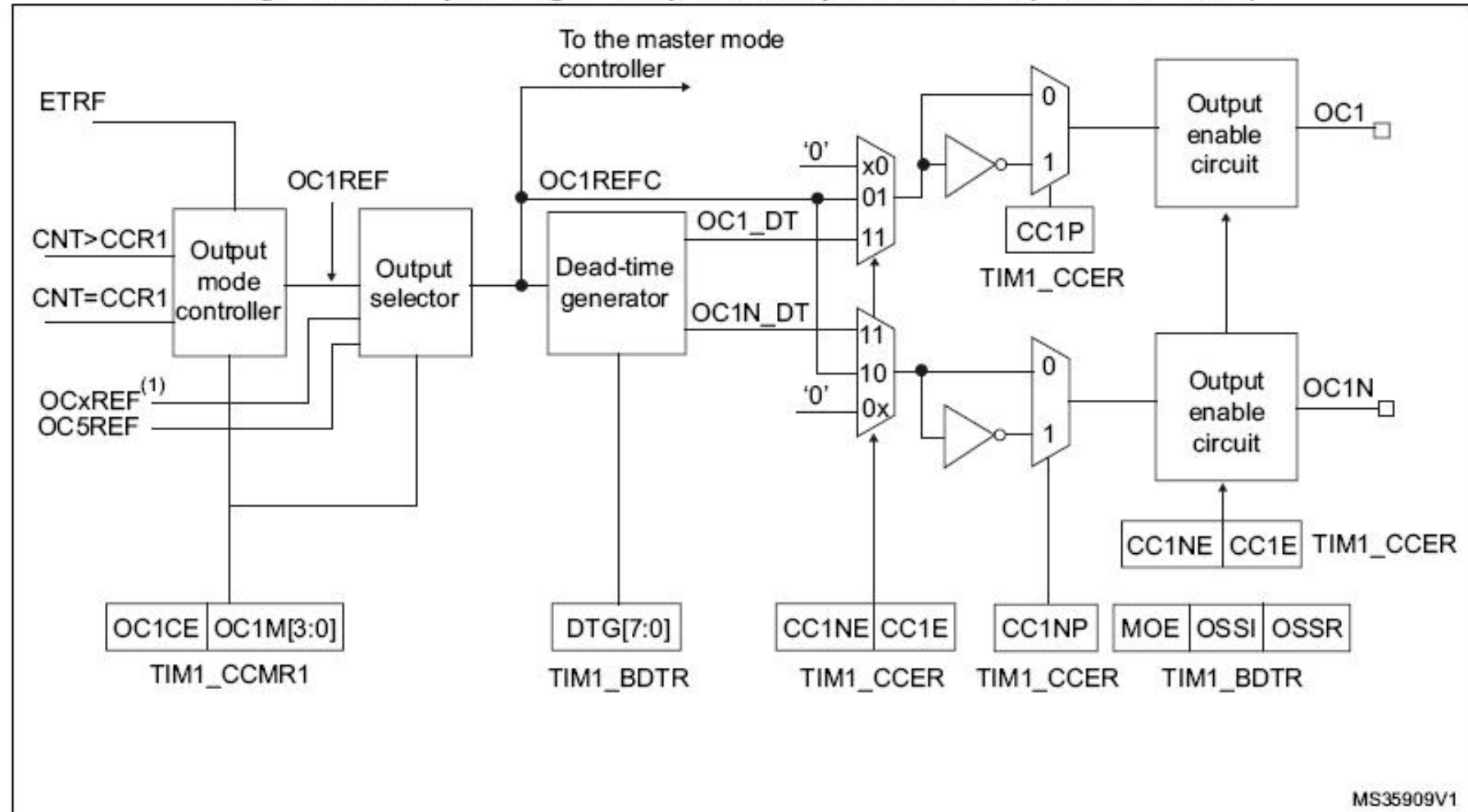


Figure 80. Output stage of capture/compare channel (channel 1 to 3)



14.3.11 Complementary outputs and dead-time insertion

The advanced-control timers (TIM1 and TIM8) can output two complementary signals and manage the switching-off and the switching-on instants of the outputs.

This time is generally known as dead-time and it has to be adjusted depending on the devices connected to the outputs and their characteristics (intrinsic delays of level-shifters, delays due to power switches...)

User can select the polarity of the outputs (main output OCx or complementary OCxN) independently for each output. This is done by writing to the CCxP and CCxNP bits in the TIMx_CCER register.

The complementary signals OCx and OCxN are activated by a combination of several control bits: the CCxE and CCxNE bits in the TIMx_CCER register and the MOE, OISx, OISxN, OSSI and OSSR bits in the TIMx_BDTR and TIMx_CR2 registers. Refer to [Table 83](#) for more details. In particular, the dead-time is activated when switching to the IDLE state (MOE falling down to 0).

Dead-time insertion is enabled by setting both CCxE and CCxNE bits, and the MOE bit if the break circuit is present. DTG[7:0] bits of the TIMx_BDTR register are used to control the dead-time generation for all channels. From a reference waveform OCxREF, it generates 2 outputs OCx and OCxN. If OCx and OCxN are active high:

- The OCx output signal is the same as the reference signal except for the rising edge, which is delayed relative to the reference rising edge.
- The OCxN output signal is the opposite of the reference signal except for the rising edge, which is delayed relative to the reference falling edge.

If the delay is greater than the width of the active output (OCx or OCxN) then the corresponding pulse is not generated.

The following figures show the relationships between the output signals of the dead-time generator and the reference signal OCxREF. (we suppose CCxP=0, CCxNP=0, MOE=1, CCxE=1 and CCxNE=1 in these examples)

14.3.12 Using the break function

When using the break function, the output enable signals and inactive levels are modified according to additional control bits (MOE, OSSI and OSSR bits in the TIMx_BDTR register, OISx and OISxN bits in the TIMx_CR2 register). In any case, the OCx and OCxN outputs cannot be set both to active level at a given time. Refer to [Table 83](#) for more details.

When a break occurs (selected level on the break input):

- The MOE bit is cleared asynchronously, putting the outputs in inactive state, idle state or in reset state (selected by the OSSI bit). This feature functions even if the MCU oscillator is off.
- Each output channel is driven with the level programmed in the OISx bit in the TIMx_CR2 register as soon as MOE=0. If OSSI=0 then the timer releases the enable output else the enable output remains high.
- When complementary outputs are used:
 - The outputs are first put in reset state inactive state (depending on the polarity). This is done asynchronously so that it works even if no clock is provided to the timer.
 - If the timer clock is still present, then the dead-time generator is reactivated in order to drive the outputs with the level programmed in the OISx and OISxN bits after a dead-time. Even in this case, OCx and OCxN cannot be driven to their active level together. Note that because of the resynchronization on MOE, the dead-time duration is a bit longer than usual (around 2 ck_tim clock cycles).
 - If OSSI=0 then the timer releases the enable outputs else the enable outputs remain or become high as soon as one of the CCxE or CCxNE bits is high.
- The break status flag (BIF bit in the TIMx_SR register) is set. An interrupt can be generated if the BIFx bit in the TIMx_BIER register is set to 1. BIFx is cleared if

```
void TIM_OCxInit(TIM_TypeDef* TIMx, TIM_OCInitTypeDef* TIM_OCInitStruct)
```

```
typedef struct
{
    uint16_t TIM_OCMode;          /*!< Specifies the TIM mode.
                                   This parameter can be a value of @ref TIM_Output_Compare_and_PWM_modes */

    uint16_t TIM_OutputState;     /*!< Specifies the TIM Output Compare state.
                                   This parameter can be a value of @ref TIM_Output_Compare_state */

    uint16_t TIM_OutputNState;    /*!< Specifies the TIM complementary Output Compare state.
                                   This parameter can be a value of @ref TIM_Output_Compare_N_state
                                   @note This parameter is valid only for TIM1 and TIM8. */

    uint16_t TIM_Pulse;           /*!< Specifies the pulse value to be loaded into the Capture Compare Register.
                                   This parameter can be a number between 0x0000 and 0xFFFF */

    uint16_t TIM_OCPolarity;      /*!< Specifies the output polarity.
                                   This parameter can be a value of @ref TIM_Output_Compare_Polarity */

    uint16_t TIM_OCNPolarity;     /*!< Specifies the complementary output polarity.
                                   This parameter can be a value of @ref TIM_Output_Compare_N_Polarity
                                   @note This parameter is valid only for TIM1 and TIM8. */

    uint16_t TIM_OCIdleState;     /*!< Specifies the TIM Output Compare pin state during Idle state.
                                   This parameter can be a value of @ref TIM_Output_Compare_Idle_State
                                   @note This parameter is valid only for TIM1 and TIM8. */

    uint16_t TIM_OCNIdleState;    /*!< Specifies the TIM Output Compare pin state during Idle state.
                                   This parameter can be a value of @ref TIM_Output_Compare_N_Idle_State
                                   @note This parameter is valid only for TIM1 and TIM8. */
} TIM_OCInitTypeDef;
```


14.4.2 TIM1 and TIM8 control register 2 (TIMx CR2)

Address offset: 0x04

Reset value: 0x0000

	15	14	13	12	11	10	9	8	
Res.	OIS4	OIS3N	OIS3	OIS2N	OIS2	OIS1N	OIS1	T	
	rw	rw	rw	rw	rw	rw	rw		

14.4.7 TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1)

Address offset: 0x18

Reset value: 0x0000

The channels can be used in input (capture mode) or in output (compare mode). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode. For a given bit, OCxx describes its function when the channel is configured in output, ICxx describes its function when the channel is configured in input. So the user must take care that the same bit can have a different meaning for the input stage and for the output stage.

[illegible]

14.4.9 TIM1 and TIM8 capture/compare enable register (TIMx_CCER)

Address offset: 0x20

Reset value: 0x0000

[illegible]

14.4.14 TIM1 and TIM8 capture/compare register 1 (TIMx_CCR1)

Address offset: 0x34

Reset value: 0x0000

[illegible]

```
void TIM_BDTRConfig(TIM_TypeDef* TIMx, TIM_BDTRInitTypeDef *TIM_BDTRInitStruct)
```

```
typedef struct
{
    uint16_t TIM_OSSRState;      /*!< Specifies the Off-State selection used in Run mode.
                                   This parameter can be a value of @ref OSSR_Off_State_Selection_for_Run_mode_state */
    uint16_t TIM_OSSIState;      /*!< Specifies the Off-State used in Idle state.
                                   This parameter can be a value of @ref OSSI_Off_State_Selection_for_Idle_mode_state */
    uint16_t TIM_LOCKLevel;      /*!< Specifies the LOCK level parameters.
                                   This parameter can be a value of @ref Lock_level */
    uint16_t TIM_DeadTime;        /*!< Specifies the delay time between the switching-off and the
                                   switching-on of the outputs.
                                   This parameter can be a number between 0x00 and 0xFF */
    uint16_t TIM_Break;          /*!< Specifies whether the TIM Break input is enabled or not.
                                   This parameter can be a value of @ref Break_Input_enable_disable */
    uint16_t TIM_BreakPolarity;  /*!< Specifies the TIM Break Input pin polarity.
                                   This parameter can be a value of @ref Break_Polarity */
    uint16_t TIM_AutomaticOutput; /*!< Specifies whether the TIM Automatic Output feature is enabled or not.
                                   This parameter can be a value of @ref TIM_AOE_Bit_Set_Reset */
} TIM_BDTRInitTypeDef;
```

14.4.18 TIM1 and TIM8 break and dead-time register (TIMx_BDTR)

Address offset: 0x44

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK[1:0]		DTG[7:0]							
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW

Note: As the bits AOE, BKP, BKE, OSSI, OSSR and DTG[7:0] can be write-locked depending on the LOCK configuration, it can be necessary to configure all of them during the first write access to the TIMx_BDTR register.

Bit 15 **MOE**: Main output enable

This bit is cleared asynchronously by hardware as soon as the break input is active. It is set by software or automatically depending on the AOE bit. It is acting only on the channels

which are configured (0: OC and OCN outputs (TIMx_CCER register). See OC/OCN enable/disable and capture/compare enable/disable).

Bit 14 **AOE**: Automatic output enable

0: MOE can be set on/off by software (not be active)
1: MOE can be set by hardware (not be active)

Note: This bit can not be modified in TIMx_BDTR register.

Bit 13 **BKP**: Break polarity

0: Break input BRK is active low
1: Break input BRK is active high

Note: This bit can not be modified in TIMx_BDTR register).

Note: Any write operation to this bit will be ignored.

Bit 12 **BKE**: Break enable

0: Break inputs (BRK and CSS) are disabled
1: Break inputs (BRK and CSS) are enabled

Note: This bit cannot be modified in TIMx_BDTR register).

Note: Any write operation to this bit will be ignored.

Bits 7:0 **DTG[7:0]**: Dead-time generator setup

This bit-field defines the duration of the dead-time inserted between the complementary outputs. DT correspond to this duration.

DTG[7:5]=0xx => DT=DTG[7:0]x t_{dtg} with $t_{dtg}=t_{DTS}$.

DTG[7:5]=10x => DT=(64+DTG[5:0])x t_{dtg} with $T_{dtg}=2xt_{DTS}$.

DTG[7:5]=110 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=8xt_{DTS}$.

DTG[7:5]=111 => DT=(32+DTG[4:0])x t_{dtg} with $T_{dtg}=16xt_{DTS}$.

Example if $T_{DTS}=125ns$ (8MHz), dead-time possible values are:

0 to 15875 ns by 125 ns steps,

16 us to 31750 ns by 250 ns steps,

32 us to 63us by 1 us steps,

64 us to 126 us by 2 us steps

Note: This bit-field can not be modified as long as LOCK level 1, 2 or 3 has been programmed (LOCK bits in TIMx_BDTR register).

Deartime ?

14.4.18 TIM1 and TIM8 break and dead-time register (TIMx_BDTR)

Address offset: 0x44

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6
MOE	AOE	BKP	BKE	OSSR	OSSI	LOCK[1:0]			
rw	rw	rw	rw	rw	rw	rw	rw		

Note: As the bits AOE, BKP, BKE, OSSR, OSSI, and the LOCK configuration, it can be needed access to the TIMx_BDTR register.

Bits 7:0 **DTG[7:0]**: Dead-time generator setup

This bit-field defines the duration of the dead-time outputs. DT correspond to this duration.

DTG[7:5]=0xx => $DT = DTG[7:0] \times t_{dtg}$ with t_{dtg}

DTG[7:5]=10x => $DT = (64 + DTG[5:0]) \times t_{dtg}$ with t_{dtg}

DTG[7:5]=110 => $DT = (32 + DTG[4:0]) \times t_{dtg}$ with t_{dtg}

DTG[7:5]=111 => $DT = (32 + DTG[4:0]) \times t_{dtg}$ with t_{dtg}

Example if $T_{DTS} = 125\text{ns}$ (8MHz), dead-time p

0 to 15875 ns by 125 ns steps,

16 us to 31750 ns by 250 ns steps,

32 us to 63us by 1 us steps,

64 us to 126 us by 2 us steps

Note: This bit-field can not be modified as long (LOCK bits in TIMx_BDTR register).

TIM_TimeBaseStructure.TIM_ClockDivision=TIM_CKD_DIV1;

14.4.1 TIM1 and TIM8 control register 1 (TIMx_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS[1:0]		DIR	OPM	URS	UDIS	CEN
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:10 Reserved, must be kept at reset value.

Bits 9:8 **CKD[1:0]**: Clock division

This bit-field indicates the division ratio between the timer clock (CK_INT) frequency and the dead-time and sampling clock (t_{DTS}) used by the dead-time generators and the digital filters (ETR, TIx),

00: $t_{DTS} = t_{CK_INT}$

01: $t_{DTS} = 2 \times t_{CK_INT}$

10: $t_{DTS} = 4 \times t_{CK_INT}$

11: Reserved, do not program this value

Advanced timer--Complementary PWM with Deadtime Insertion and Break

1、 Enable timer clock

```
RCC_APB2PeriphClockCmd();
```

2、 Timer Clock Configuration

CK_INT , default setting ,
can be omitted

```
TIM_DeInit(); TIM_InternalClockConfig();
```

3、 Initialization the time base structure

```
TIM_TimeBaseInit();
```

4、 Initialization the output compare structure

```
TIM_OCxInit();
```

5、 Initialization the BDTR structure

```
TIM_BDTRConfig();
```

6、 Enable the Preload of TIMx-CCR& ARR

```
TIM_OCxPreloadConfig();  
TIM_ARRPreloadConfig();
```

Optional

7、 Enable MOE , Enable timer

```
TIM_CtrlPWMOutputs();  
TIM_Cmd();
```

```
TIM_GPIO_Config()  
GPIO_Mode_AF_PP
```

0x4001 3C00 - 0x4001 3FFF	ADC3	APB2
0x4001 3800 - 0x4001 3BFF	USART1	
0x4001 3400 - 0x4001 37FF	TIM8 timer	
0x4001 3000 - 0x4001 33FF	SPI1	
0x4001 2C00 - 0x4001 2FFF	TIM1 timer	
0x4001 2800 - 0x4001 2BFF	ADC2	
0x4001 2400 - 0x4001 27FF	ADC1	
0x4001 2000 - 0x4001 23FF	GPIO Port G	
0x4001 1C00 - 0x4001 1FFF	GPIO Port F	
0x4001 1800 - 0x4001 1BFF	GPIO Port E	
0x4001 1400 - 0x4001 17FF	GPIO Port D	
0x4001 1000 - 0x4001 13FF	GPIO Port C	

Table 83. Output control bits for complementary OCx and OCxN channels with break feature

Control bits					Output states ⁽¹⁾	
MOE bit	OSSI bit	OSSR bit	CCxE bit	CCxNE bit	OCx output state	OCxN output state
1	X	0	0	0	Output Disabled (not driven by the timer), OCx=0, OCx_EN=0	Output Disabled (not driven by the timer), OCxN=0, OCxN_EN=0
		0	0	1	Output Disabled (not driven by the timer), OCx=0, OCx_EN=0	OCxREF + Polarity OCxN=OCxREF xor CCxNP, OCxN_EN=1
		0	1	0	OCxREF + Polarity OCx=OCxREF xor CCxP, OCx_EN=1	Output Disabled (not driven by the timer) OCxN=0, OCxN_EN=0
		0	1	1	OCREF + Polarity + dead-time OCx_EN=1	Complementary to OCREF (not OCREF) + Polarity + dead-time OCxN_EN=1
		1	0	0	Output Disabled (not driven by the timer) OCx=CCxP, OCx_EN=0	Output Disabled (not driven by the timer) OCxN=CCxNP, OCxN_EN=0
		1	0	1	Off-State (output enabled with inactive state) OCx=CCxP, OCx_EN=1	OCxREF + Polarity OCxN=OCxREF xor CCxNP, OCxN_EN=1
		1	1	0	OCxREF + Polarity OCx=OCxREF xor CCxP, OCx_EN=1	Off-State (output enabled with inactive state) OCxN=CCxNP, OCxN_EN=1
		1	1	1	OCREF + Polarity + dead-time OCx_EN=1	Complementary to OCREF (not OCREF) + Polarity + dead-time OCxN_EN=1
0	0	X	0	0	Output Disabled (not driven by the timer) Asynchronously: OCx=CCxP, OCx_EN=0, OCxN=CCxNP, OCxN_EN=0 Then if the clock is present: OCx=OISx and OCxN=OISxN after a dead-time, assuming that OISx and OISxN do not correspond to OCX and OCxN both in active state.	
	0		0	1		
	0		1	0		
	0		1	1		
	1		0	0	Off-State (output enabled with inactive state) Asynchronously: OCx=CCxP, OCx_EN=1, OCxN=CCxNP, OCxN_EN=1 Then if the clock is present: OCx=OISx and OCxN=OISxN after a dead-time, assuming that OISx and OISxN do not correspond to OCX and OCxN both in active state	
	1		0	1		
	1		1	0		
	1		1	1		

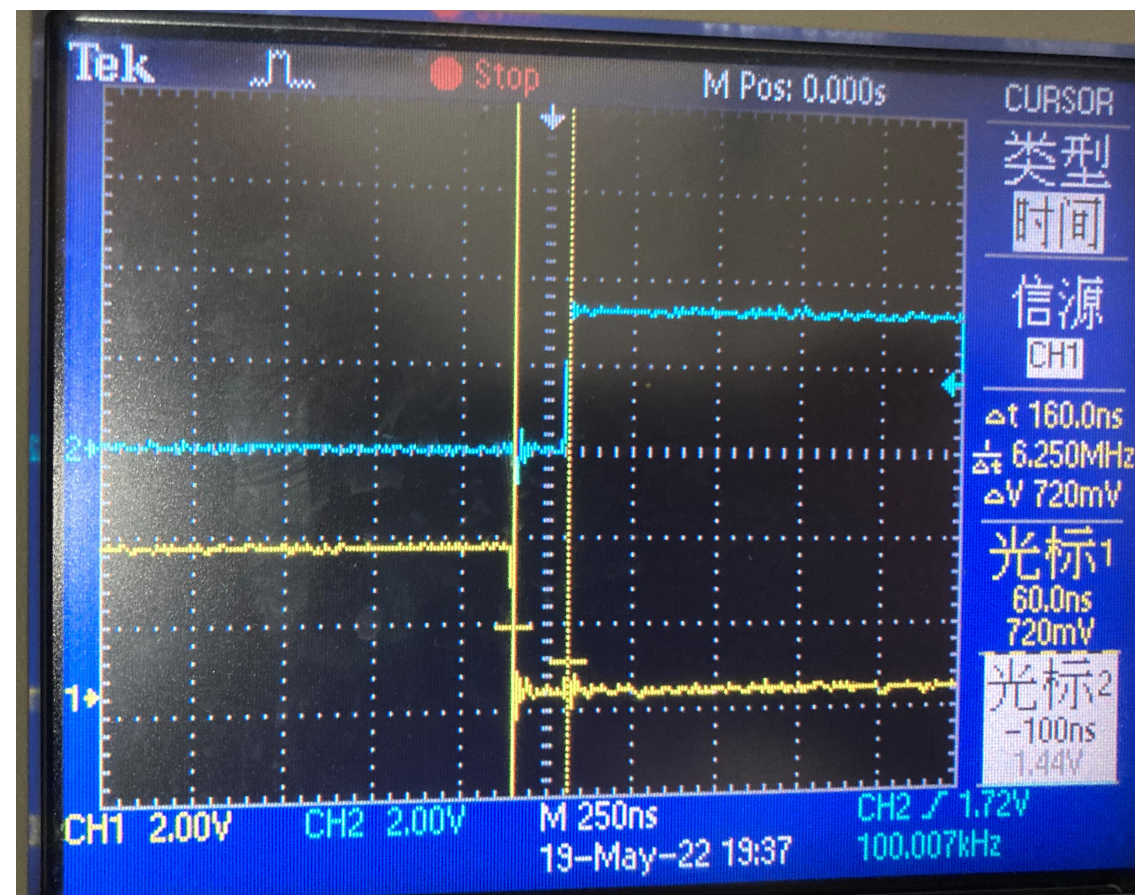
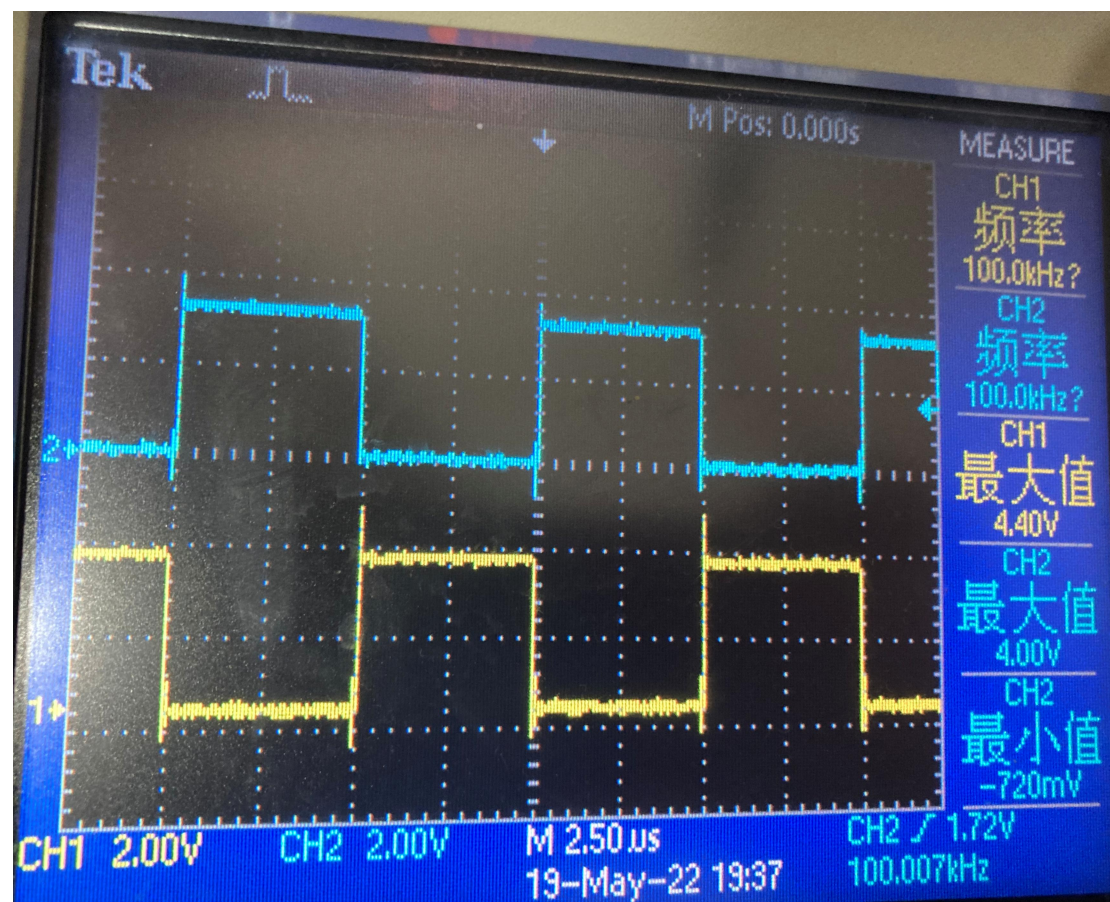


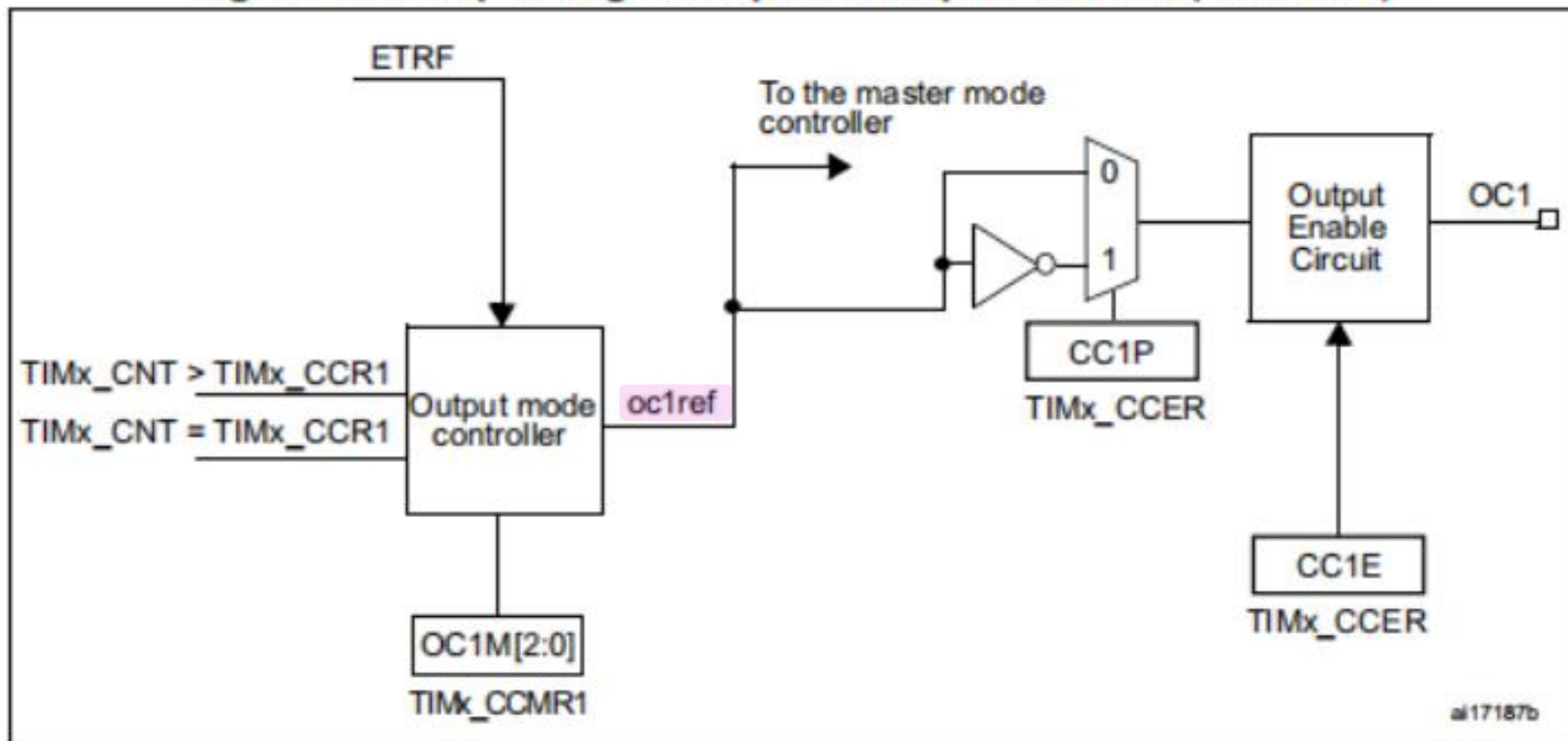
表 32-1 高级控制和通用定时器通道引脚分布

	高级定时器		通用定时器			
	TIM1	TIM8	TIM2	TIM5	TIM3	TIM4
CH1	PA8/PE9	PC6	PA0/PA15	PA0	PA6/PC6/PB4	PB6/PD12
CH1N	PB13/PA7/PE8	PA7				
CH2	PA9/PE11	PC7	PA1/PB3	PA1	PA7/PC7/PB5	PB7/PD13
CH2N	PB14/PB0/PE10	PB0				
CH3	PA10/PE13	PC8	PA2/PB10	PA2	PB0/PC8	PB8/PD14
CH3N	PB15/PB1/PE12	PB1				
CH4	PA11/PE14	PC9	PA3/PB11	PA3	PB1/PC9	PB9/PD15
ETR	PA12/PE7	PA0	PA0/PA15		PD2	PE0
BKIN	PB12/PA6/PE15	PA6				

Homework1

- Write out the registers those are used(configurated or read) by the function **TIM_OCInit()** & **TIM_BDTRInit()** and their meanings one by one;
- Modify the frequency and duty of the output PWM ,
Verify its break function based on the project “高级定时器-输出PWM带死区” ----Easy frequency = 150K , duty= 50% ,

Figure 127. Output stage of capture/compare channel (channel 1)



General TIM

